Table 4
We use GloVe [32] as the word vector representation. We use PARARULES with all depths as the training set for all models and then test them on examples with different reasoning depths (D). Comparison among our IMA-GloVe-GA, IMA-GloVe, MAC-GloVe, DMN-GloVe, IMASM-GloVe, LSTM-GloVe, and RoBERTa-Large on PARARULES test sets with different reasoning depths.

Train ↓; Test →	D=1	D=2	D=3	D≤3	D≤3+NatLang	D≤5	D≤5+NatLang
IMA-GloVe	0.861	0.853	0.830	0.842	0.810	0.792	0.705
MAC-GloVe	0.792	0.776	0.750	0.763	0.737	0.701	0.652
DMN-GloVe	0.846	0.843	0.817	0.827	0.789	0.779	0.666
IMASM-GloVe	0.864	0.855	0.824	0.838	0.801	0.782	0.608
LSTM-GloVe	0.500	0.500	0.500	0.499	0.499	0.500	0.500
IMA-GloVe-GA	**0.950**	**0.943**	**0.919**	**0.927**	**0.883**	**0.879**	**0.741**
RoBERTa-Large	**0.986**	**0.985**	**0.977**	**0.979**	**0.972**	**0.967**	**0.949**

Adam[2] is 1e-02. The latent dimension d for GRU is set to 64. The loss function is binary cross entropy because the task is a binary classification problem. The evaluation measure is accuracy. It is computed as $\text{accuracy} = n_{\text{correct}}/n_{\text{total}}$, where n_{correct} is the number of correctly classified examples and n_{total} is the total number of examples. We evaluate the Transformer baseline model (RoBERTa-Large) from FAIRSEQ [40] that was included the original paper that introduced PARARULES [3]. We follow the official script[3] to fine-tune the model, we use the same hyperparameter in [40] to fine-tune RoBERTa-Large on PARARULES. We use the initial learning rate of 1e-05, and we use 16 as the batch size. We conduct all experiments using the NVIDIA 460.84 Linux Driver. The CPU version is Intel(R) Xeon(R) Gold 5218 CPU @ 2.30GHz and 16 cores. The CUDA version is 11.2 and a Quadro RTX 8000 with 48 GB GPU memory is used for our experiments.

Experimental Results: Table 4 shows the results of the RNN-based models on test sets that require different reasoning steps. Unless otherwise stated models are trained on the entire training set with all reasoning depths. The first horizontal row denotes test sets with specific reasoning depth(s), and the second horizontal row denotes the number of test samples. We find that our IMA-GloVe-GA and RoBERTa-Large achieve the 2nd and 1st best results. These two models also achieve the 2nd and 1st best results in the test set with additional human-rewritten examples (+NatLang), showing a better generalisation performance and robustness. We find that by adding gated attention the test accuracy improved over IMA-GloVe in all cases. The results support that gated attention can be more effective than dot-product attention on examples that require multi-step reasoning. IMA-GloVe and IMASM-GloVe obtained better performance than the other RNN-based baseline models. We speculate that the dot-product attention enhances the learning and representation between context and question. In addition, the vanilla LSTM fails to converge in all those cases. We get similar results as reported in DeepLogic [1] that the vanilla GRU/LSTM failed on the multi-step reasoning over logic programs. A possible reason is that the vanilla LSTM is more sensitive to hyperparameter tuning. It is harder to train to converge than the other models.

[2]Keras Optimizers, https://keras.io/api/optimizers/
[3]FAIRSEQ, https://github.com/pytorch/fairseq

Table 5
IMA-GloVe, IMA-GloVe-GA, and RoBERTa-Large trained on CONCEPTRULES V1 (simplified / full) and tested on different test sets. Rules in CONCEPTRULES V1 Simplified are not shuffled, while CONCEPTRULES V1 full contains randomly shuffled rules. CONCEPTRULES V1 full has larger number of relations and entities than CONCEPTRULES V1 simplified.

Model	Train set	Test accuracy (Simplified Test set)	Test accuracy (Full Test set)
IMA-GloVe	Simplified	0.994	0.729
	Full	0.844	**0.997**
IMA-GloVe-GA	Simplified	**0.998**	**0.747**
	Full	0.851	**0.999**
RoBERTa-Large	Simplified	**0.997**	0.503
	Full	**0.927**	0.995

The results shown in Table 4 compare the performances of the IMA models based on the GloVe word embeddings and the pretrained RoBERTa model on the same test sets of different reasoning depths. We find that by adding gated attention the test accuracy improved over IMA-GloVe in all cases. The results support that gated attention can be more effective than dot-product attention on examples that require multi-step reasoning. We also find that the pretrained RoBERTa model achieves better results in all of the cases. A possible reason is that our model is trained from scratch without any large-scale pretraining. However, our model can perform better than the other baselines without any pretraining.

Table 5 shows the results on CONCEPTRULES V1 (simplified) and CONCEPTRULES V1 (full). The second column is the training set CONCEPTRULES V1 (simplified or full). The last two columns show test accuracy on CONCEPTRULES V1 (simplified or full) test set. For example, in the first row, we train IMA-GloVe on the simplified version of the training sets, and then test on different test sets. We find that IMA-GloVe-GA (with gated attention) performs better than IMA-GloVe in all cases. In contrast, the pretrained RoBERTa model only achieves high performance in the simplified test set when it is trained on the simplified training set. It indicates that the pretrained RoBERTa overfits CONCEPTRULES V1 (simplified) since the rules are not shuffled. Most likely it learns spurious relations in the training data and fails to generalise on test examples with shuffled rules.

Table 6 shows results on CONCEPTRULES V2. We select the four models used in Table 5, and respectively select data of different depths in CONCEPTRULES V2 (full) as the training data. For example, Mod3 represents a model trained on examples with depth of 3. Mod0123 represents a model trained on examples with depth of 3 and less. Each row of the table represents a test set of a specific reasoning depth. We find that the IMA-based model achieves better results in almost all cases. The pretrained RoBERTa model does not outperform the IMA-based models in all cases. We find that when the model is only trained on examples with reasoning step of 1, the test accuracy dropped as we increase the reasoning steps in the test set. However, the drop in performance is not as great as we expected for the IMA-based models. This shows that the IMA-based models generalise better on out-of-distribution test examples. A possible reason why models achieve higher test accuracies on CONCEPTRULES V2 (full) is that CONCEPTRULES V2 has much more training data compared to PARARULES, from Table 2.

Table 6
IMA-GloVe, IMA-GloVe-GA, and RoBERTa-Large trained on CONCEPTRULES V2 (full) and tested on test sets that require different depths of reasoning.

Model	Test set	Mod1 Depth=1	Mod2 Depth=2	Mod3 Depth=3	Mod01 Depth≤1	Mod012 Depth≤2	Mod0123 Depth≤3
IMA-GloVe	Depth=1	**0.999**	**0.998**	**0.990**	**0.997**	**0.998**	**0.997**
	Depth=2	**0.998**	**0.999**	**0.988**	**0.995**	**0.998**	**0.997**
	Depth=3	**0.997**	0.998	0.981	**0.991**	0.996	**0.997**
IMA-GloVe-GA	Depth=1	0.993	0.996	0.987	0.987	0.991	**0.997**
	Depth=2	0.993	**0.999**	0.974	0.986	0.991	0.995
	Depth=3	0.988	**1**	**0.994**	0.989	**0.997**	0.994
RoBERTa-Large	Depth=1	0.998	0.975	0.831	0.995	0.975	0.971
	Depth=2	0.997	0.972	0.885	0.993	0.972	0.965
	Depth=3	0.987	0.951	0.984	0.988	0.951	0.936

Table 7
RoBERTa-Large trained on PARARULES with different reasoning depths and tested on test sets that require different depths of reasoning. A bold number indicates the highest accuracy in a test set.

Model	Test set	Mod012 (Depth≤2)	Mod0123 (Depth≤3)	Mod0123Nat (Depth≤3+NatLang)	Mod012345 (Depth≤5)
RoBERTa-Large	Depth=0	**0.971**	0.946	0.968	0.953
	Depth=1	**0.943**	0.907	0.933	0.909
	Depth=2	**0.933**	0.902	0.932	0.902
	Depth=3	0.562	0.902	**0.926**	0.907
	Depth=4	0.481	0.863	**0.904**	0.888
	Depth=5	0.452	0.856	0.916	**0.933**
	NatLang	0.573	0.579	**0.962**	0.594

Table 7 shows the results of fine-tuning RoBERTa-Large on datasets with different reasoning depths and testing on test sets of various reasoning depths. Depth≤2 means the model is trained on the dataset with depths no larger than 2. Depth≤3+NatLang represents the model that is trained on the dataset with depths no larger than 3 and extra examples paraphrase by humans. The models trained on the datasets with shallow depths have lower test accuracies on test sets that require deeper reasoning depths. We find that adding training examples that require deeper reasoning steps for Mod0123 improves the results on the test sets with deeper reasoning steps (e.g. Mod0123Nat vs. Mod0123, and Mod012345 vs. Mod0123). Additionally, we find that adding the examples paraphrased by humans improves the performance on test examples with deep reasoning steps and human-paraphrased test examples.

Table 8 shows the results of fine-tuning RoBERTa-Large on the datasets found in Table 7 and our PARARULE-Plus. The addition of PARARULE-Plus during the fine-tuning improves the performance on the examples that require more reasoning steps. The yellow background shows the improvement over the test accuracy reported in Table 7, and the bracket contains the magnitude of the improvement. The bold numbers indicate the highest test accuracy on corresponding test sets. The results support that our PARARULE-Plus addresses the depth